

Member ID: _____

Time: _____

Rank: _____



MANAGERIAL ACCOUNTING (135)

STATE 2022

CONCEPT KNOWLEDGE:

Multiple Choice (15 @ 2 points each) _____ (30 points)

APPLICATION KNOWLEDGE:

Short Answer _____ (87 points)

Problem 1: Production Cost Report _____ (68 points)

Problem 2: Budgeting Scenarios _____ (45 points)

TOTAL POINTS _____ ***(230 points)***

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

EXAM GUIDELINES:

You have been hired as a Financial Assistant and will be keeping the accounting records for Digital Solutions, located at 700 Morse Road, Suite 201, Columbus, Ohio 43214. Digital Solutions provides accounting and other financial services for clients. You will complete jobs for Digital Solutions' own accounting records, as well as for clients.

You will have 90 minutes to complete your work. The test is divided into two parts: concept knowledge and application knowledge.

Your name and/or school name should *not* appear on any work you submit for grading. Write your Member ID in the provided space. Staple all pages in order before you turn in your test.

Assumptions to make when taking this assessment:

- Round all calculations to two decimal places at the final step.
- Round all percentages to one decimal place.
- Use 360 days for interest calculations.

Multiple Choice

1. Which of the following companies is most likely to use process costing?
 - a. A breakfast cereal company
 - b. An accounting firm
 - c. A building contractor
 - d. A music studio

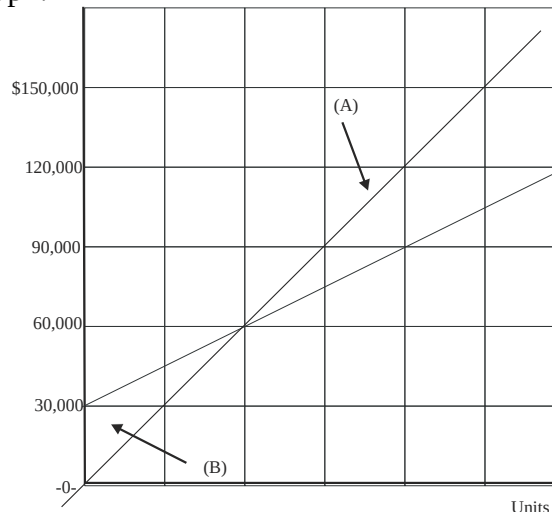
2. In a process costing system,
 - a. Work-in-process inventory employs one general ledger account with a subsidiary ledger.
 - b. Product costs are expensed as incurred
 - c. Separate work-in-process inventory accounts are maintained for each process or department.
 - d. A job cost record is maintained for each job.

3. Which of the following best describes the term equivalent units?
 - a. Different types of units that have same selling price
 - b. Substitute of units that are partially completed
 - c. Partially completed units of output that will be sold as is
 - d. Partially completed units expressed in terms of fully completed units

4. _____ are costs incurred to avoid poor-quality goods or services.
 - a. Appraisal costs
 - b. External failure costs
 - c. Prevention costs
 - d. Internal failure costs

5. Stalcup Cafe operates a chain of coffee shops. The company pays rent of \$24,000 per year for each shop. Supplies (napkins, bags and condiments) are purchased as needed. The managers of each shop are paid a salary of \$3,000 per month and all other employees are paid on an hourly basis. The cost of rent is which kind of cost?
 - a. Fixed cost
 - b. Variable cost
 - c. Mixed cost
 - d. Relevant cost

6. Consider the following graph:



The line designated by the letter (A) represents which of the following?

- a. Variable cost
 - b. Breakeven
 - c. Total revenue
 - d. Total cost
7. Management of Schumacher Corporation is considering whether to purchase a new model 370 machine costing \$464,000 or a new model 220 machine costing \$405,000 to replace a machine that was purchased 10 years ago for \$439,000. The old machine was used to make product I43L until it broke down last week. Unfortunately, the old machine cannot be repaired. Management has decided to buy the new model 220 machine. It has less capacity than the new model 370 machine, but its capacity is sufficient to continue making product I43L. Management also considered, but rejected, the alternative of simply dropping product I43L. If that were done, instead of investing \$340,000 in the new machine, the money could be invested in a project that would return a total of \$456,000. In making the decision to buy the model 220 machine rather than the model 370 machine, the sunk cost was _____.
- a. \$405,000
 - b. \$456,000
 - c. \$464,000
 - d. \$439,000

8. The budgeting process that involves adding a month to the end of the budget period at the end of each month in order to maintain a twelve-month planning horizon is referred to as _____.
a. participative budgeting
b. continuous budgeting
c. capital budgeting
d. monthly budgeting
9. When the sales price per unit decreases, the breakeven point _____.
a. remains the same
b. increases
c. decreases
d. decreases proportionately
10. When the total fixed costs increase, the breakeven point _____.
a. decreases
b. remains the same
c. increases
d. decreases proportionately
11. The starting point in developing the master budget is the preparation of the _____.
a. production budget
b. sales budget
c. budgeted income statement
d. cash budget
12. Burch Co. normally produces between 50,000 and 75,000 units each year. Producing more than 75,000 units alters the company's cost structure. For example, fixed costs increase because more space must be rented and additional supervisors must be hired. The production range between 50,000 and 75,000 is called the _____.
a. differential range
b. relevant range
c. opportunity range
d. leverage range
13. DiSimoni, Inc. is preparing financial statements to be distributed to investors and creditors. The company should prepare the income statement using _____.
a. variable costing because it is better for planning purposes
b. variable costing because it follows GAAP
c. absorption costing because it is better for controlling purposes
d. absorption costing because it follows GAAP
14. Elza Corp. incurs both fixed and variable production costs. Assuming that production is within the relevant range, if volume goes up by 20%, then the total costs would _____.

- a. remain the same
- b. decrease by 20%
- c. increase by 20%
- d. increase by an amount less than 20%

15. Which of the following is considered a period cost under variable costing but not under absorption costing?

- a. Fixed manufacturing overhead
- b. Variable selling and administrative costs
- c. Fixed selling and administrative costs
- d. Variable manufacturing costs

Short Answer (87 Points)

1. O'Brien Corporation, a merchandising company, reported the following results for October:

Sales	\$413,000
Cost of goods sold (all variable)	\$173,100
Total variable selling expense	\$23,000
Total fixed selling expense	\$19,300
Total variable administrative expense	\$17,400
Total fixed administrative expense	\$31,000

What is the contribution margin for October?

Contribution Margin for October	
---------------------------------	--

2. Saunders Corporation produces and sells a single product. Data concerning that product appear below:

	Per Unit	Percent of Sales
Selling price	\$ 140	100 %
Variable expenses	84	60 %
Contribution margin	\$ <u>56</u>	<u>40</u> %

The company is currently selling 5,300 units per month. Fixed expenses are \$185,000 per month. The marketing manager believes that a \$5,300 increase in the monthly advertising budget would result in a 200 unit increase in monthly sales. What should be the overall effect on the company's monthly net operating income of this change?

_____ of \$ _____
 Increase/decrease Amount

3. McNew Inc. produces and sells a single product. The selling price of the product is \$200.00 per unit and its variable cost is \$40.00 per unit. The fixed expense is \$407,820 per month. Round the units to the correct whole number and the dollars to the nearest cent.

Break-even point in units	
Break-even point in sales dollars	
How many units does McNew need to sell to earn \$250,000 in profit?	

4. Bobb Corporation's contribution format income statement for July appears below:

Sales	\$ 273,600
Variable expenses	<u>123,120</u>
Contribution margin	150,480
Fixed expenses	<u>49,660</u>
Net operating income	<u>\$ 100,820</u>

What is the degree of operating leverage?

Degree of Operating Leverage	
------------------------------	--

5. Spangler Inc. has an operating leverage of 4.8. If the company's sales increase by 13%, its net operating income should increase by what percentage?

Percentage increase in operating income	
---	--

6. The Refining Department of Weber, Inc. had 63,000 tons of sugar to account for in December. Of the 63,000 tons, 45,000 tons were completed and transferred to the Boiling Department, and the remaining 18,000 tons were 60% complete with respect to conversion costs. The materials required for production are added at the beginning of the process. Conversion costs are added equally throughout the refining process. The weighted-average method is used.

What are the total equivalent units of production for direct materials?	
What are the total equivalent units of production for conversion costs?	

7. Hendrickson Corporation has provided the following information:

	Cost per Unit	Cost per Period
Direct materials	\$6.95	
Direct labor	\$4.00	
Variable manufacturing overhead	\$1.45	
Fixed manufacturing overhead		\$109,800
Sales commissions	\$1.05	
Variable administrative expense	\$0.75	
Fixed selling and administrative expense		\$40,050

What is the incremental manufacturing cost that the company will incur if it increases production from 9000 to 9001 units?

Incremental Manufacturing Cost	
--------------------------------	--

8. Hewes Inc. produces and sells two products. Data concerning those products for the most recent month appear below:

	Product DE	Product AH
Sales	\$ 51,000	\$ 56,000
Variable expenses	\$ 13,900	\$ 26,760

The fixed expenses of the entire company were \$39,050.

What is the break-even point in sales dollars for the entire company?	
---	--

What is the break-even point in sales dollars for each product?

Product DE	
Product AH	

9. Martin Co.'s phone bill consists of both fixed and variable costs. Refer to the four-month data below and apply the high-low method to answer the question.

	Minutes	Total Bill
January	460	\$3,025
February	240	\$2,675
March	210	\$2,500
April	320	\$2,840

What is the variable cost per minute?	
What is the fixed portion of the total cost?	
If 390 minutes are used in May, how much will the total bill be?	

10. Schultz Company has the following data for the Assembly Department for August:

Units in process at the beginning of August	800
Units started in August	2,000
Units completed and transferred	2,100
Units in process at end of August	700

Conversion costs are added evenly throughout the process. The company uses the weighted-average method. Compute the equivalent units of production for direct materials and conversion costs for each independent scenario:

	Direct Materials	Conversion Costs
Units in process at the end of August are 20% complete; materials are added at the beginning of the process.		
Units in process at the end of August are 80% complete; materials are added at the beginning of the process.		
Units in process at the end of August are 30% complete; materials are added at the end of the process.		
Units in process at the end of August are 75% complete; materials are added at the halfway point.		

11. Nelson & Lott Co. incurred the following costs while producing 500 units: direct materials, \$10 per unit; direct labor, \$25 per unit; variable manufacturing overhead, \$15 per unit; total fixed overhead costs, \$10,000; variable selling and administrative costs, \$5 per unit; total fixed selling and administrative costs, \$7,500. There are no beginning inventories.

What is the unit product cost using variable costing?	
What is the unit product cost using absorption costing?	
What is the operating income using absorption costing if 450 units are sold for \$100 each?	
What is the operating income using variable costing if 450 units are sold for \$100 each?	

Problem 1 (68 Points)

Robsplach Co. prepares and packages fertilizer for commercial landscaping business. Robsplach has two departments: Mixing and Packaging. Direct materials are added at the beginning of the mixing process and at the end of the packaging process. Conversion costs are added evenly throughout each process. Data from the month of May for the Mixing Department are as follows:

Gallons		Costs	
Beginning Work-in-Process Inventory	0 gallons	Beginning Work-in-Process Inventory	\$0
Started in production	8,500 gallons	Costs added during May:	
Completed and transferred out to Packaging in May	6,000 gallons	Direct materials	\$5,525
Ending Work-in-Process Inventory (10% of the way through the blending process)	2,500 gallons	Direct labor	1,500
		Manufacturing overhead allocated	<u>2,125</u>
		Total costs added during May	<u>\$ 9,150</u>

Prepare the Production Cost Report for the Mixing Department on the next page by filling in all the unshaded box.

Robsplach Co.				
Production Cost Report – Mixing Dept				
Month Ended May 31, 2022				
		Equivalent Units		
Units	Whole Units	Direct Materials	Conversion Costs	
Units to account for:				
Beg WIP				
Started into Production				
Total units to account for				
Units accounted for				
Completed & transferred out				
End WIP				
Total units accounted for				
Costs		Direct Materials	Conversion Costs	Total Costs
Costs to account for				
Beg WIP				
Costs added during period				
Total costs to account for				
Divided by Total EUP				
Cost per EUP				
Costs accounted for				
Completed & transferred out				
End WIP				
Total costs accounted for				

Problem 2 (45 Points)

Each of the following budget scenarios should be answered independently.

A) Thomas Company makes a product that is very popular as a Christmas gift. Thus, peak sales occur in December of each year, as shown in the sales budget for the fourth quarter given below:

	October	November	December	Total
Budgeted Sales (all on account)	\$200,000	\$300,000	\$500,000	\$1,000,000

From past experience, the company has learned that 20% of a month's sales are collected in the month of sale, 70% are collected in the month following sale, and the remaining 10% are collected in the second month following sale. Bad debts are negligible and can be ignored.

How much cash is expected to be collected in December?	
What will be the balance in Accounts Receivable at the end of December?	

B) Swenson Company has budgeted sales of its popular moose harness for the next four months as follows:

	Unit Sales
January	50,000
February	75,000
March	90,000
April	80,000

The company is now in the process of preparing a production budget for the first quarter. Past experience has shown that end-of-month inventory levels must equal 10% of the following month's unit sales. The inventory at the end of December was 5,000 units.

How many units should the company plan to produce in January?	
---	--

C) Three grams of rose oil are required for each bottle of PBA, a very popular perfume made by a small company in Ohio. The cost of rose oil is \$1.50 per gram. Budgeted production of PBA is given below for the last quarter of Year 2 and the first quarter of Year 3.

	Year 2			Year 3
	October	November	December	January
Budgeted Production (in bottles)	60,000	90,000	150,000	70,000

The inventory of rose oil at the end of the month must be equal to 20% of the following month's production needs. Approximately 12,000 grams of rose oil will be on hand at the start of October.

How many grams of rose oil should be purchased on October?	
How much will the rose oil cost for the October purchases?	
The company pays for 60% of the purchases in the month of purchase, and 40% in the month following purchase. Accounts Payable at the end of September is \$90,000. How much will the cash payments be for October?	
What will be the balance in Accounts Payable at the end of October?	

D) Lowe Inc. expects to produce 8,000 units in the next quarter. Each unit requires 0.35 direct labor hours, and each employee is paid \$12.00 per hour.

How many hours of direct labor should Lowe budget for?	
How much direct labor cost should Lowe budget for?	